I^{2}C Communication Interface

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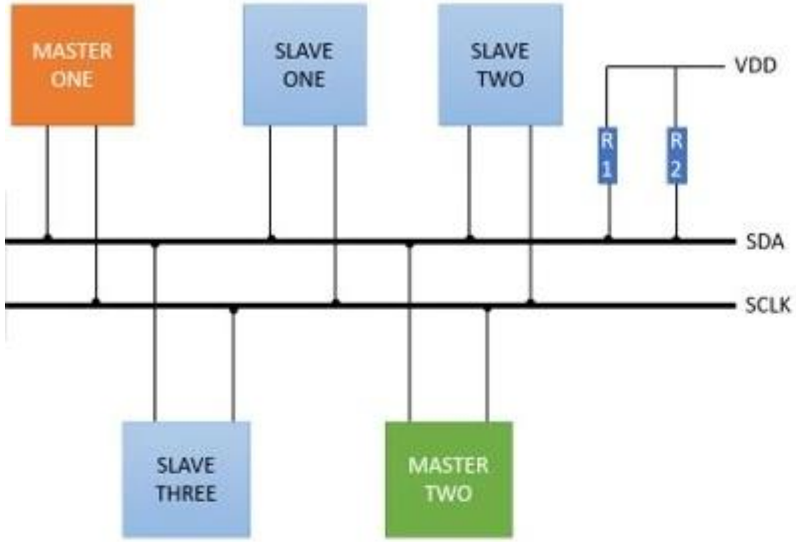
I2C Communication Interface

Clock and Data Lines:

- **SCL (Serial Clock)** - Always controlled by master
- **SDA (Serial Data)** - Bidirectional, controlled by master or slave depending on operation

Acknowledgment:

- **ACK** - Receiver pulls SDA low during 9th clock pulse
- **NACK** - Receiver leaves SDA high during 9th clock pulse
- Master sends NACK after receiving the last byte in a read operation



Multi-byte Operations:

- Multiple data bytes can be sent/received in a single transaction
- Each byte requires its own ACK/NACK

Address Format:

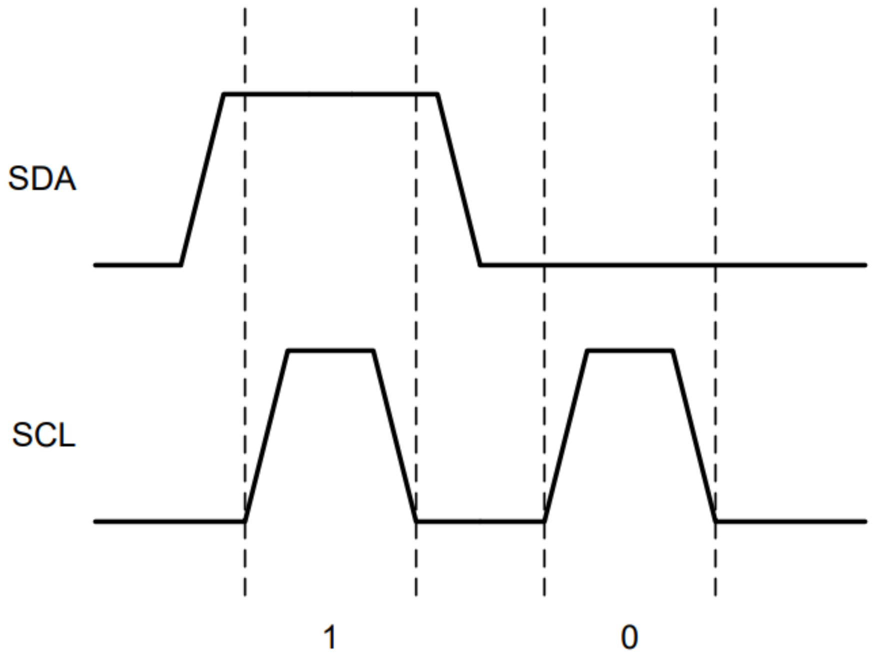
- 7-bit addressing is most common
- 10-bit addressing available for extended address space
- Read/Write bit is the LSB of the address byte

Multi-Master Operations:

- I2C supports multiple devices to work as a master on the same bus
- Arbitration mechanism is used to avoid bus conflict between multiple masters

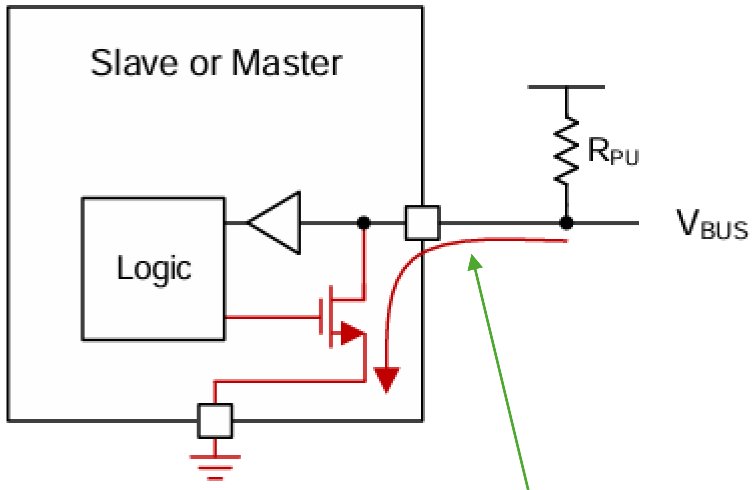
I^2C Digital One and Zero Representations

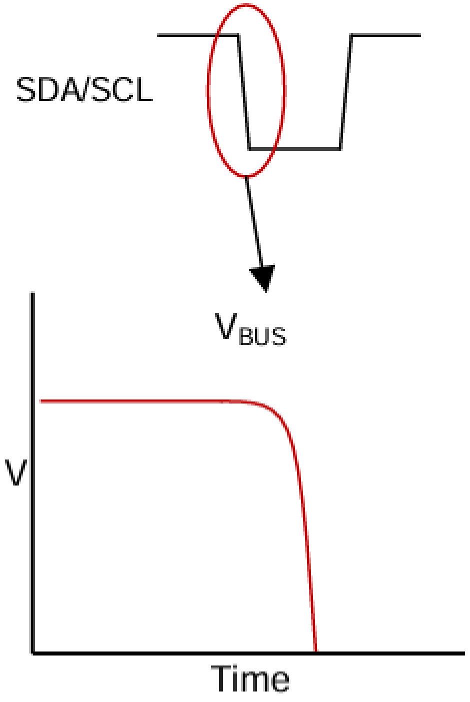
- A logical one is sent when the SDA releases the line, allowing the pullup resistor to pull the line to a high level.
- A logical zero is sent when SDA pulls down on the line, setting a low level near ground.
- For a valid bit, SDA does not change between a rising edge and the falling edge of SCK for that bit. Changes of the SDA between the rising and falling edges of the SCL can be interpreted as a START or STOP condition on the I2C bus.



Pulling the Bus Low With An Open-Drain Interface

When the slave or master wishes to transmit a logic low, it will activate the FET. The device pulls current through the resistor to ground. This pulls the open-drain line low.

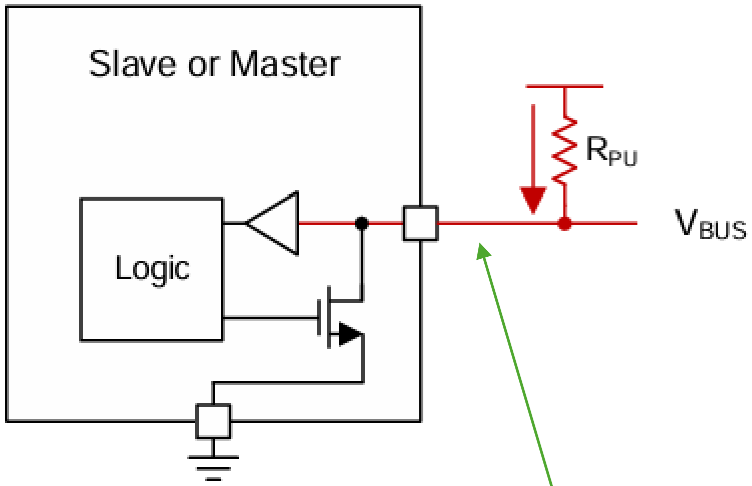


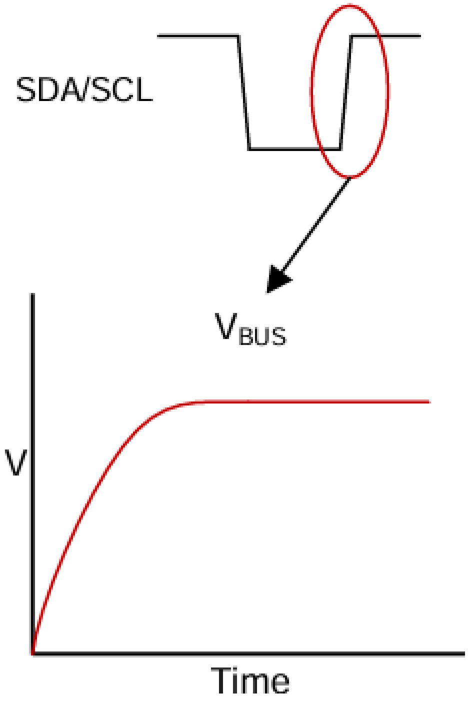


The flow of current to pull the bus
LOW

Releasing the Bus With An Open-Drain Interface

When the slave or master wishes to transmit a logic high, it may only release the bus by turning off the pull-down FET. This leaves the bus floating, and the pull-up resistor will pull the voltage up to the voltage rail, which will be interpreted as a high.

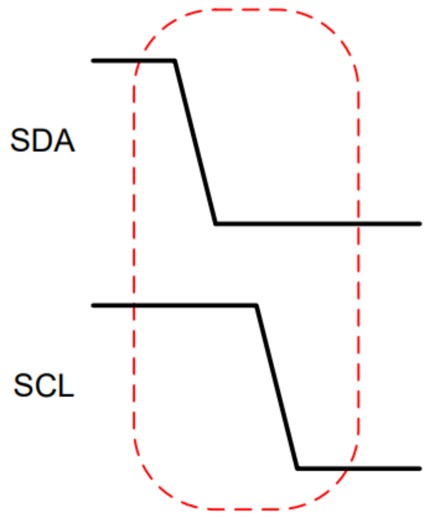




The flow of current to pull the bus
HIGH

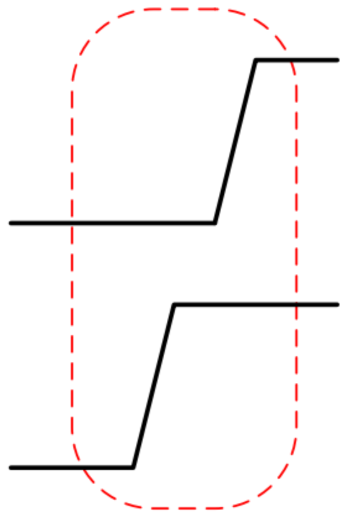
I^{2}C START and STOP Conditions

- **I2C communication with a device is initiated by the master sending a START condition and terminated by the master sending a STOP condition.**
- **START Condition: A high-to-low transition on the SDA line while the SCL is high**
- **STOP Condition: A low-to-high transition on the SDA line while the SCL is high**



I^{2}C START

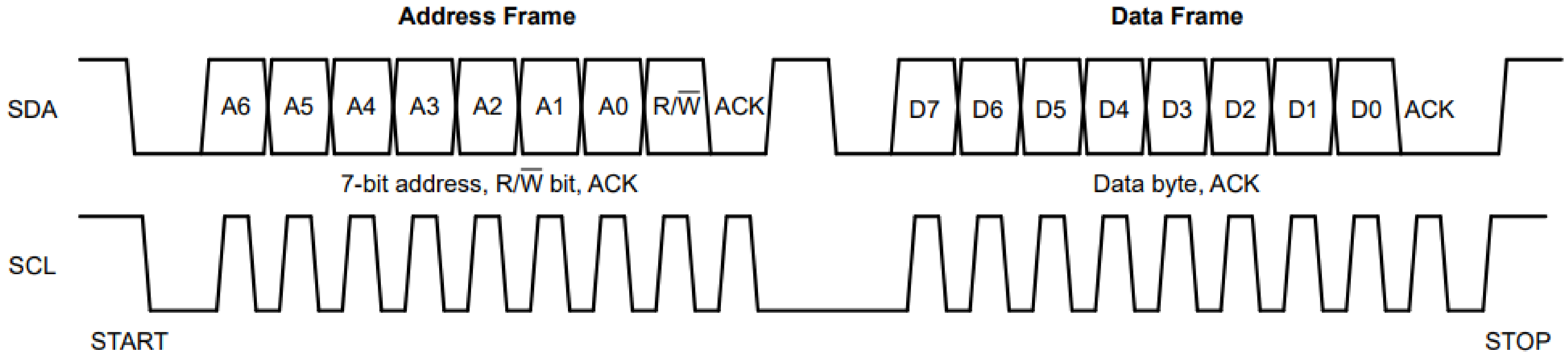
A controller device claims the I^{2}C bus for communication with a target device



I^{2}C STOP

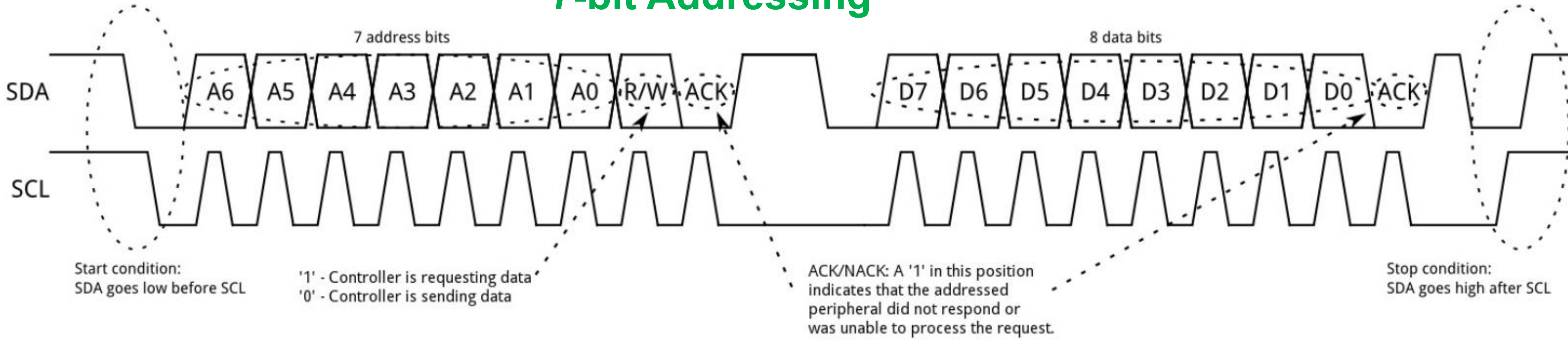
A controller device completes communication with a target device and releases the I^{2}C bus

I^{2}C Address and Data Frames

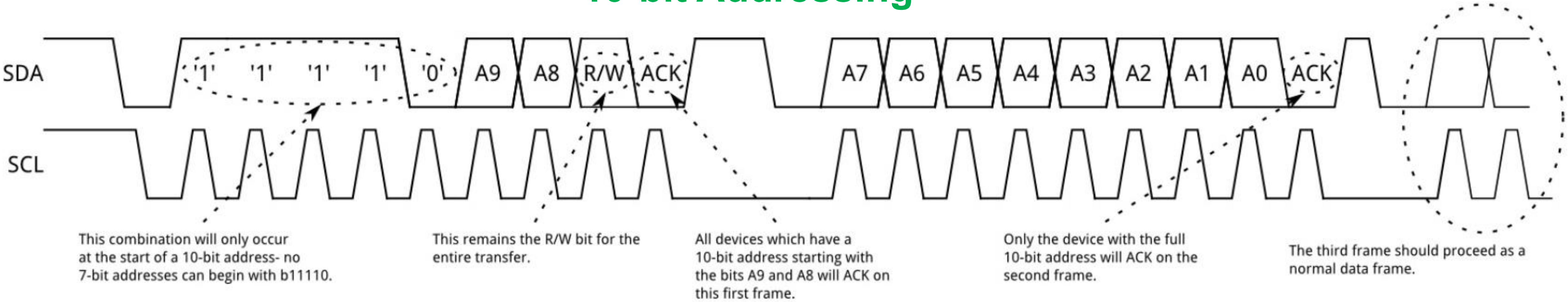


Addressing in I²C

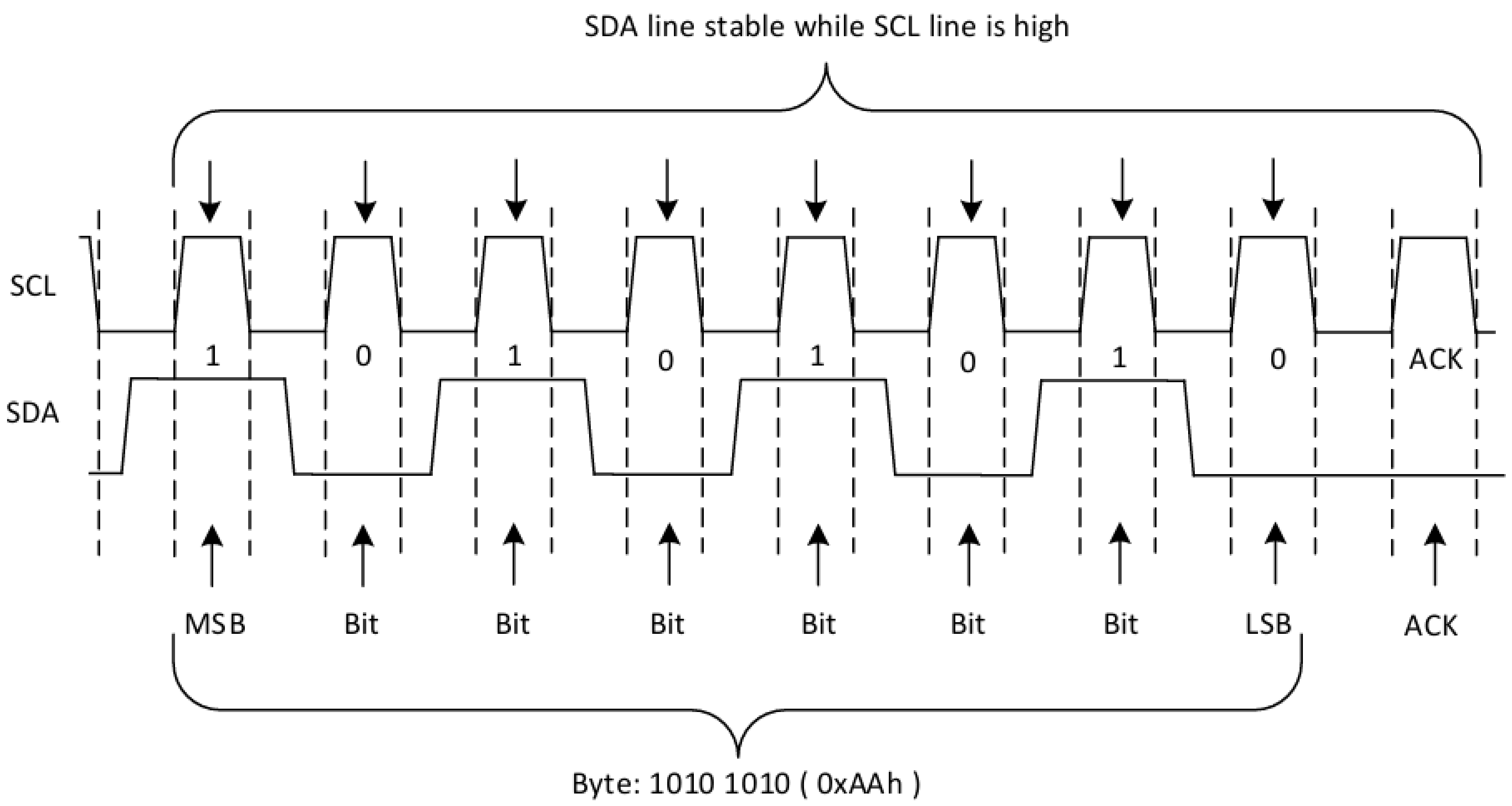
7-bit Addressing



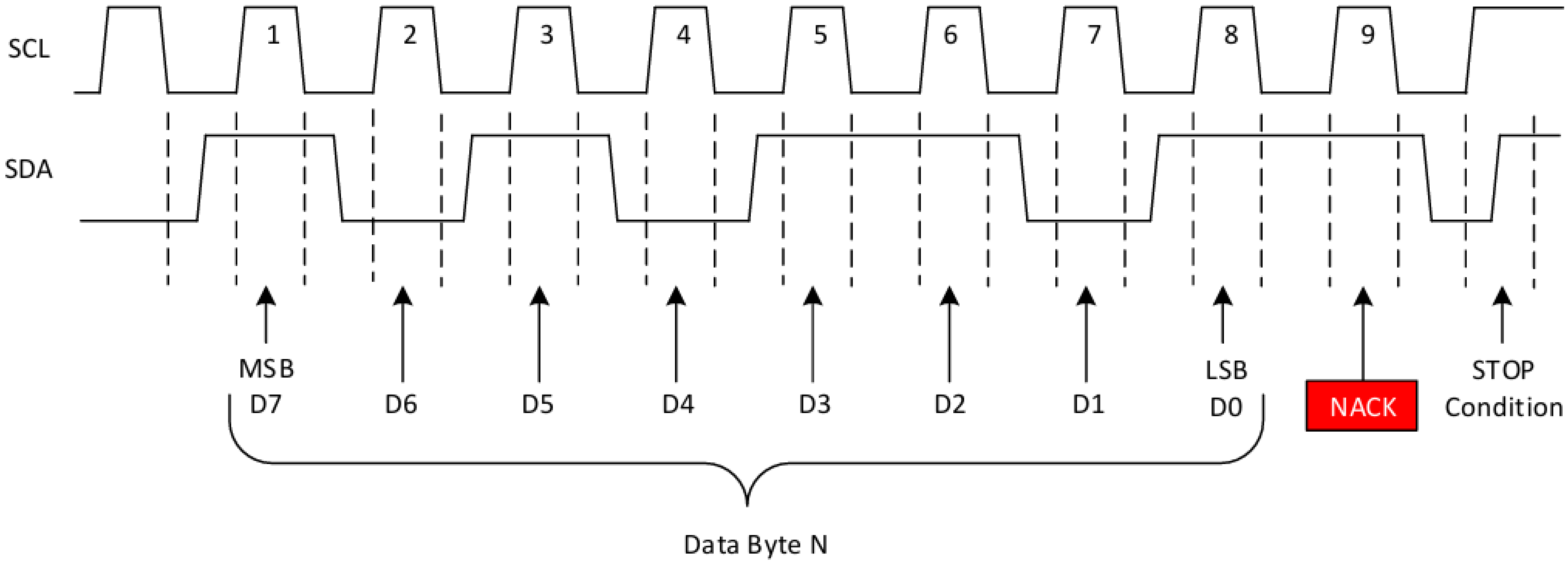
10-bit Addressing



Single Byte Data Transfer in I2C



Acknowledge (ACK) and Not Acknowledge (NACK)



Example of NACK Waveform

Advantages and Disadvantages

Advantages:

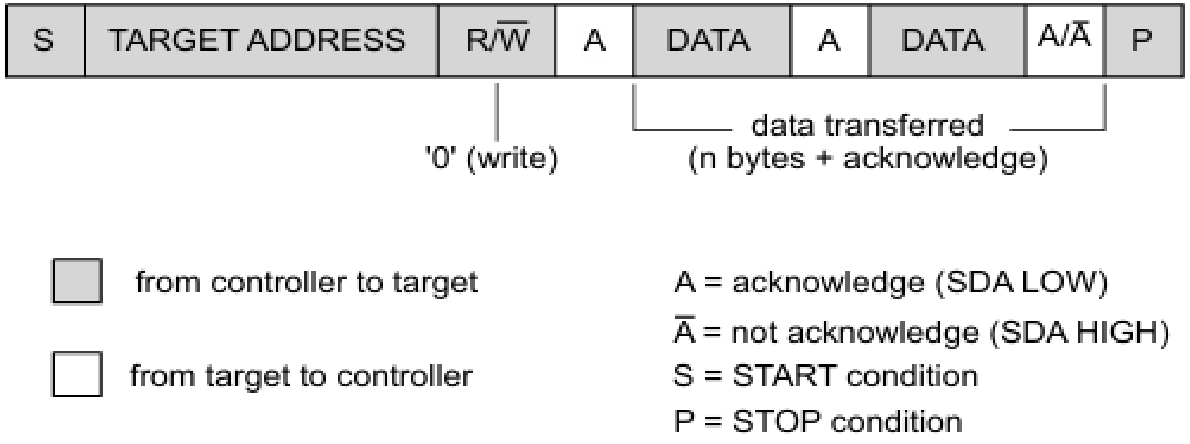
- Multiple processors can share same I2C bus
- Cost-effective (no additional hardware needed)
- Automatic conflict resolution

Disadvantages:

- **Increased complexity** in software
- **Potential delays** due to arbitration and retries
- **Bus efficiency** decreases with more masters
- **Debugging difficulty** - intermittent conflicts

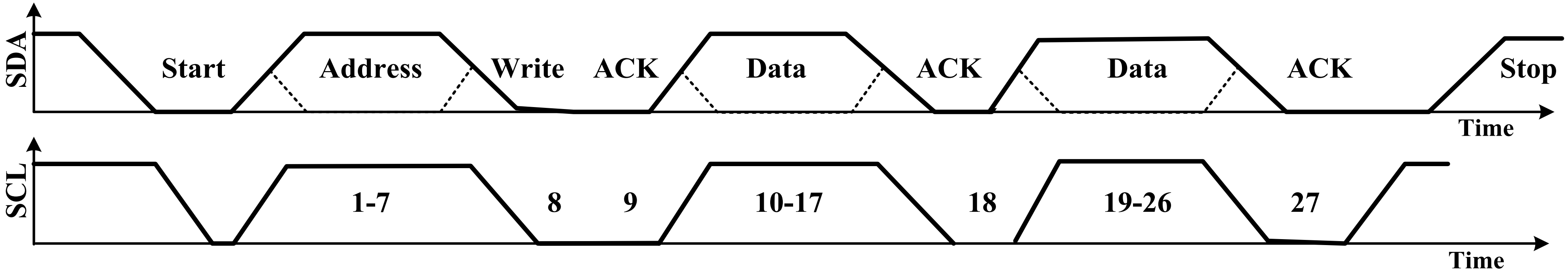
Practice Example # 1

Draw the timing waveforms for the following data transfer structure using a 7-bit addressing I2C protocol. **Compute** the amount of data being transferred, data transfer rate, and frame rate. The operating frequency is 100 kHz.



Solution to Practice Example # 1

The timing waveform:



The total amount of bits transferred = Target address bits + Read/Write bit + Acknowledgment bit + Data bit + Acknowledgment bit + Data bit + Acknowledgment bit = 7 + 1 + 1 + 8 + 1 + 8 + 1 = 27 bits

Operating frequency, f = 100 kHz.

Clock period, $T = 1/f$ = 1/100kHz = 0.01 ms = 10 us.

The total time for data transfer of 27 bits = 27×10 ms = 270 ms.

270 ms time is required to transfer 27 bits of data

$\therefore$ 1 ms time is required to transfer $\frac{27}{270}$ bits of data

$\therefore$ 1 s = 10^6 ms time is required to transfer $\frac{27 \times 10^6}{270}$ bits of data

That is, 10^5 bits of data are transferred per second, or 100 kbps.

Solution to Practice Example # 1 (Contd...)

The total amount of bits transferred in one frame = Target address bits + Read/Write bit + Acknowledgment bit + Data bit + Acknowledgment bit = 7 + 1 + 1 + 8 + 1 = 18 bits

The total time for data transfer of 1 frame, i.e., 18 bits = 18×10 = 180 ms.

180 ms time is required to transfer 1 frame of data

$\therefore$ 1 ms time is required to transfer $\frac{1}{180}$ frame

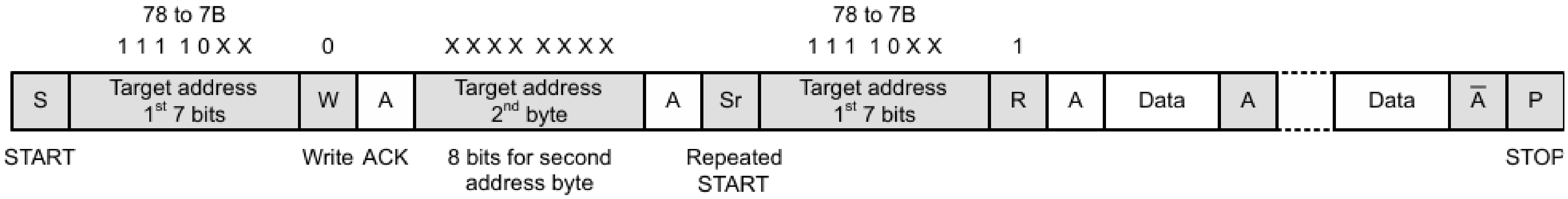
$\therefore$ 1 s = 10^6 ms time is required to transfer $\frac{1\times10^6}{180}$ frame

That is, $\frac{10^5}{18} = 5555.55$ frames are transferred each second.

Therefore, the frame rate is 5.55 kfps.

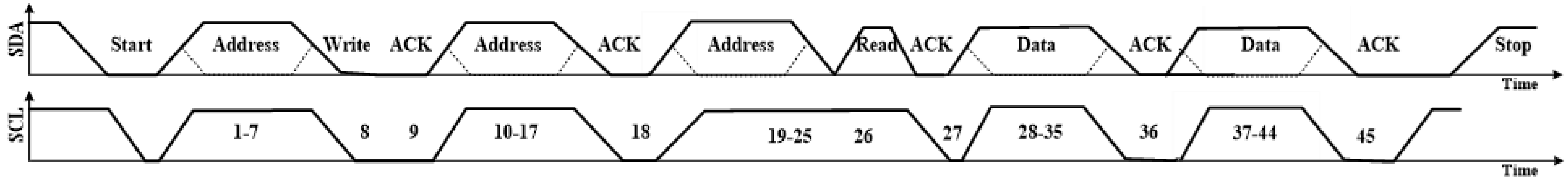
Practice Example # 2

Draw the timing waveforms for the following data transfer structure using a 10-bit addressing I2C protocol. **Compute** the amount of data being transferred, data transfer rate, and frame rate. The operating frequency is 5 MHz.



Solution to Practice Example # 2

The timing waveform is given below:



The total amount of bits transferred = Target address bits + Read/Write bit + Acknowledgment bit + Target address bits + Acknowledgment bit + Target address bits + Read/Write bit + Acknowledgment bit + Data bit + Acknowledgment bit + Data bit + Acknowledgment bit = 7 + 1 + 1 + 8 + 1 + 7 + 1 + 1 + 8 + 1 + 8 + 1 = 45 bits

The operating frequency, f = 5 MHz. So, the clock period, $T = 1/f$ = 1/5MHz = 0.2 ms

The total time for data transfer of 45 bits = 45×0.2 = 9 ms.

9 ms time is required to transfer 45 bits of data

$\therefore$ 1 ms time is required to transfer $\frac{45}{9}$ bits of data

$\therefore$ 1 s = 10^6 ms time is required to transfer $\frac{45\times10^6}{9}$ bits of data

That is, 5×10^6 bits of data are transferred per second, or 5 Mbps.

Solution to Practice Example # 2 (Contd...)

The total amount of bits transferred in one frame = Target address bits + Read/Write bit + Acknowledgment bit + Target address bits + Acknowledgment bit + Data bit + Acknowledgment bit = 7 + 1 + 1 + 8 + 1 + 8 + 1 = 27 bits

The total time for data transfer of 1 frame, i.e., 27 bits = 27×0.2 = 5.4 ms.

5.4 ms time is required to transfer 1 frame of data

$\therefore$ 1 ms time is required to transfer $\frac{1}{5.4}$ frame

$\therefore$ 1 s = 10^6 ms time is required to transfer $\frac{1\times10^6}{5.4}$ frame

That is, $\frac{10^6}{5.4} = 185185$ frames are transferred each second. Therefore, the frame rate is 185.185 kfps.